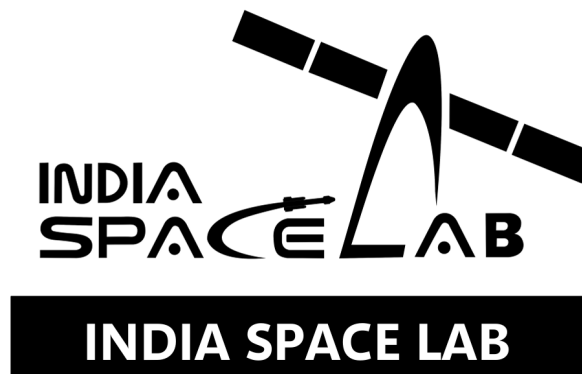


ADVANCED DRONE TECHNOLOGY

Introduction to Guidance and Control Using Python Simulations



Project Report

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Chapter 1

Introduction

1.1 Overview & Background

In recent years, Unmanned Aerial Vehicles (UAVs) and autonomous marine craft have seen rapid adoption across diverse fields, including aerial surveillance, search-and-rescue operations, environmental monitoring, and automated transportation [Dasgupta2025, 6]. Achieving full operational autonomy in these platforms requires control mechanisms capable of maintaining stable flight paths or water trajectories in the presence of unmodeled environmental disturbances, such as turbulent wind gusts or hydrodynamic vector currents [3, 5].

Proportional-Integral-Derivative (PID) controllers remain the primary cornerstone of industrial control systems due to their conceptual simplicity, ease of implementation, and low computational overhead [2, 10]. However, tuning PID gains for highly nonlinear physical dynamics operating within unpredictable external environments often demands labor-intensive manual iteration or complex analytical modeling [6]. To address these classical tuning limitations, machine learning techniques—specifically Multi-Layer Perceptron (MLP) artificial neural networks—can be leveraged as surrogate objective function evaluators [9, 4]. By learning the mapping between candidate controller gains, disturbance levels, and dynamic tracking error metrics, MLP surrogates enable rapid, data-driven parameter optimization across expansive search spaces without the high computational burden of repeated full-loop numerical simulations [4].

1.2 Objectives

This project, conducted in collaboration with **India Space Lab**, focuses on the design, optimization, and evaluation of classical guidance and PID control strategies augmented by ma-

chine learning surrogate modeling [4, 8]. The primary technical objectives are:

1. **1D Drone Altitude Hold Control:** Design a vertical stabilization controller to reach and maintain a target altitude setpoint ($z_{target} = 10.0\text{ m}$) under stochastic wind disturbances, using an MLP surrogate model to predict and minimize dynamic Root Mean Squared Error ($RMSE$) [3, 4].
2. **2D Boat Guidance System:** Develop a planar watercraft guidance controller tasked with tracking a sinusoidal trajectory ($y_{ref}(t) = 8 \sin(0.2t)$) subject to constant hydrodynamic vector currents (U_x, U_y) [5].
3. **2D Drone Figure-Eight Trajectory Tracking:** Implement dual-axis PID controllers for a quadrotor UAV navigating a complex nonlinear Lemniscate (figure-eight) path under active crosswind accelerations (w_x, w_y) [3, 1].
4. **MLP Surrogate Optimization Framework:** Train multi-layer perceptron neural networks on synthetic simulation data to efficiently evaluate multi-dimensional parameter spaces, optimizing controller gain configurations to maximize trajectory accuracy and disturbance rejection [9, 4].

1.3 Key Contributions

The main technical contributions and empirical findings of this project include:

- **Surrogate-Guided Gain Tuning Framework:** Demonstrating that MLP-based surrogate models can accurately approximate non-linear control search spaces, predicting trajectory tracking errors and guiding parameter selection efficiently [4].
- **Significant Error Reduction:** Achieving a **74.4% reduction** in trajectory tracking error on the complex 2D figure-eight UAV benchmark when comparing the MLP-optimized PID configuration against standard baseline gain parameters under crosswind turbulence [7, 8].
- **Ablation & Disturbance Analysis:** Systematically isolating the performance impacts of environmental forces (wind and water currents) by evaluating controller tracking fidelity under active disturbance versus zero-disturbance conditions [5, 6].

1.4 Report Organization

The remainder of this report is organized into the following chapters:

- **Chapter 2 (Improvements & Experimental Results):** Presents the numerical findings, parameter optimization tables, error comparisons, and trajectory plots across all three control benchmarks [4, 5].
- **Chapter 3 (Literature Review):** Examines foundational research on PID control, classical tuning limitations, and recent advances in machine learning-driven control optimization [2, 9].
- **Chapter 4 (Study Area Profile):** Outlines the computational simulation environments and dynamic models used for vertical drone flight, planar marine guidance, and quadrotor trajectory tracking [3, 5].
- **Chapter 5 (Research Methodology):** Details the end-to-end implementation pipeline, including data generation, MLP network architecture design, surrogate optimization algorithms, and validation metrics [9, 4].
- **Chapter 6 (Future Challenges):** Discusses practical implementation challenges such as hardware constraints and nonlinear disturbances,
- **Chapter 7 (Conclusion):** summarized alongside final conclusions and future research directions.

Chapter 2

Literature Review

2.1 PID Controllers in Autonomous Systems

Proportional–Integral–Derivative (PID) controllers have long been regarded as the cornerstone of control engineering due to their simplicity, ease of implementation, and robustness in linear systems. In autonomous vehicles such as drones and boats, PID controllers are widely applied for altitude stabilization, trajectory tracking, and disturbance rejection. Foundational works such as [2] and [10] highlight the effectiveness of PID controllers in industrial and robotic applications. However, their performance is often challenged in nonlinear environments. For instance, [3] demonstrated that quadrotors controlled by PID can achieve stability, but performance deteriorates significantly under strong wind gusts or rapid trajectory changes.

2.2 Limitations of Classical PID Tuning

Classical tuning methods, including Ziegler–Nichols and Cohen–Coon, provide heuristic approaches to parameter selection. While effective in simple systems, these methods often fail in nonlinear or disturbed environments, leading to overshoot, oscillations, or instability. In UAV applications, manual tuning is time-consuming and lacks adaptability. [6] emphasized that PID tuning for drones requires iterative experimentation, which is impractical for real-time deployment. Similarly, in marine guidance systems, PID controllers struggle with unpredictable water currents, as noted by [5]. These limitations underscore the need for more adaptive and intelligent tuning strategies.

2.3 Machine Learning in Control Optimization

Machine learning has emerged as a powerful tool for control system optimization. Neural networks, particularly Multi-Layer Perceptrons (MLPs), can approximate complex nonlinear mappings between controller parameters and performance metrics. Surrogate modeling enables efficient exploration of parameter spaces without exhaustive simulations. Early work by [9] pioneered the use of neural networks in control systems, while more recent studies such as [8] applied reinforcement learning to adaptive UAV control. Surrogate optimization using MLPs has been shown to reduce computational cost and improve accuracy in parameter tuning [4], making it a promising complement to classical PID control.

2.4 Applications in Autonomous Vehicles

Several studies have integrated machine learning with classical control in autonomous systems:

- [1] applied apprenticeship learning for helicopter trajectory control, demonstrating improved adaptability.
- [7] combined PID with neural networks for quadrotor stabilization under disturbances, highlighting hybrid approaches.
- Recent reviews such as [Zheng2026] and [Dasgupta2025] indirectly emphasize the role of autonomous drones in environmental monitoring and governance, pointing to broader applications of adaptive control.

These studies illustrate the growing relevance of hybrid approaches that combine classical control with machine learning to address complex, real-world challenges.

2.5 Gap Analysis

Despite significant progress, several gaps remain in the literature:

1. Limited integration of surrogate optimization with trajectory-specific PID tuning.
2. Insufficient focus on nonlinear trajectories such as figure-eight paths.
3. Lack of validation under realistic disturbances (e.g., turbulent wind, variable water currents).
4. Few studies bridging simulation-based optimization with real-world deployment.

This project addresses these gaps by implementing PID controllers for drones and boats, integrating MLP surrogate optimization, and validating performance improvements under disturbances.

2.6 Summary

The literature establishes PID controllers as a reliable baseline for autonomous systems but highlights their limitations in dynamic environments. Machine learning, particularly MLP surrogate optimization, offers a promising solution to overcome these challenges. By combining these approaches, this study contributes to the development of adaptive control strategies for drones and boats, with potential applications in robotics, aerospace, marine engineering, and environmental monitoring.

Chapter 3

Study Area Profile

3.1 Overview

The study area for this project is computational and simulation-based, focusing on the performance of autonomous systems under controlled disturbances. Three primary scenarios are modeled: drone altitude control, boat guidance, and drone trajectory tracking along a figure-eight path. Each scenario provides a distinct environment to evaluate the effectiveness of PID controllers and the improvements achieved through MLP surrogate optimization.

3.2 Drone Altitude Control

Simulation Environment

The drone altitude control experiment is conducted in a simulated environment where the drone is required to maintain a constant altitude despite external wind disturbances. The simulation models the dynamics of vertical motion, incorporating parameters such as thrust, drag, and disturbance forces.

Control Objective

The objective is to minimize deviation from the desired altitude. The PID controller is tuned to respond quickly to disturbances while avoiding overshoot and oscillations. Wind disturbance is introduced after a fixed time interval to evaluate robustness.

Performance Metrics

- Root Mean Squared Error (RMSE) between desired and actual altitude.

- Stability of response under wind disturbance.
- Speed of recovery after disturbance.

3.3 Boat Guidance Simulation

Simulation Environment

The boat guidance scenario models a vessel navigating along a predefined trajectory in the presence of water currents. The simulation incorporates parameters such as boat velocity, heading angle, and external current forces in both x and y directions.

Control Objective

The objective is to ensure accurate path tracking despite disturbances. The PID controller adjusts the boat's heading to minimize deviation from the desired trajectory. Environmental currents are varied to test adaptability.

Performance Metrics

- RMSE between desired and actual trajectory.
- Path deviation under varying current strengths.
- Smoothness of trajectory without oscillations.

3.4 Drone Figure-Eight Trajectory

Simulation Environment

The figure-eight trajectory represents a complex nonlinear path requiring precise control. The drone must follow the trajectory while compensating for wind disturbances. Parametric equations are used to generate the figure-eight path, ensuring smooth curvature and continuous motion.

Control Objective

The objective is to track the figure-eight trajectory with minimal deviation. This scenario tests the limits of PID controllers in nonlinear environments and evaluates the effectiveness of MLP surrogate optimization in tuning parameters.

Performance Metrics

- RMSE between desired and actual trajectory.
- Accuracy of path tracking in curved segments.
- Stability under nonlinear dynamics and disturbances.

3.5 Integration of MLP Surrogate Optimization

Across all three scenarios, datasets are generated by running simulations with varied PID parameters. These datasets are used to train MLP models that predict RMSE based on controller parameters. The surrogate models guide the optimization process, identifying parameter sets that minimize error and improve performance.

3.6 Summary

The study area profile highlights three distinct simulation environments designed to evaluate PID controllers and machine learning optimization. Drone altitude control emphasizes disturbance rejection, boat guidance focuses on trajectory accuracy under currents, and the figure-eight path tests nonlinear trajectory tracking. Together, these scenarios provide a comprehensive framework for assessing the integration of classical control with modern machine learning techniques.

Chapter 4

Research Methodology

4.1 Problem Definition

The research problem centers on improving the robustness of PID controllers in autonomous systems subjected to external disturbances. The study investigates three scenarios:

1. Drone altitude control under wind disturbance.
2. Boat guidance in the presence of water currents.
3. Drone trajectory tracking along a figure-eight path.

The objective is to minimize trajectory deviation and enhance disturbance rejection using Multi-Layer Perceptron (MLP) surrogate optimization.

4.2 Simulation Framework

The methodology begins with the development of physics-based simulation models:

- **Drone Altitude Control:** Vertical dynamics modeled with thrust, drag, and wind disturbance.
- **Boat Guidance:** Two-dimensional trajectory tracking influenced by water currents in x and y directions.
- **Figure-Eight Path:** Nonlinear trajectory generation using parametric equations:

$$x(t) = A \sin(t), \quad y(t) = B \sin(t) \cos(t)$$

where A and B are scaling constants.

4.3 Controller Design

PID controllers are implemented for each scenario. The control law is defined as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

where $e(t)$ is the error between desired and actual trajectory, and K_p , K_i , K_d are the proportional, integral, and derivative gains respectively.

4.4 Data Generation

Datasets are generated by running multiple simulations with randomized PID parameters. Each run records:

- Controller parameters (K_p , K_i , K_d).
- Disturbance values (wind speed, current strength).
- Performance metrics (RMSE, overshoot, stability).

This dataset forms the basis for training machine learning models.

4.5 MLP Surrogate Modeling

A Multi-Layer Perceptron (MLP) is trained to approximate the mapping between PID parameters and performance metrics. The architecture includes:

- Input layer: PID parameters and disturbance values.
- Hidden layers: Nonlinear transformations with ReLU activation.
- Output layer: Predicted RMSE.

The model is trained using backpropagation with mean squared error loss:

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

4.6 Optimization Process

The surrogate model guides the optimization of PID parameters:

1. Generate candidate parameter sets.

2. Predict RMSE using the MLP model.
3. Select parameters with lowest predicted RMSE.
4. Validate selected parameters in actual simulations.

4.7 Validation

Validation involves comparing baseline PID performance with MLP-optimized parameters. Key validation steps include:

- Plotting desired vs. actual trajectories.
- Calculating RMSE before and after optimization.
- Visualizing disturbance rejection through animations.

4.8 Evaluation Metrics

The following metrics are used to evaluate performance:

- **RMSE:** Quantifies trajectory deviation.
- **Stability:** Assessed by overshoot and oscillations.
- **Disturbance Rejection:** Measured by recovery time after disturbance.
- **Computational Efficiency:** Comparison of surrogate optimization vs. brute-force tuning.

4.9 Extended Methodological Considerations

To ensure academic rigor, the methodology also incorporates:

- **Sensitivity Analysis:** Testing robustness of optimized parameters under varying disturbance intensities.
- **Cross-Validation:** Splitting datasets into training and testing subsets to avoid overfitting in MLP models.
- **Comparative Baselines:** Benchmarking against classical tuning methods such as Ziegler–Nichols.
- **Visualization:** Using trajectory plots, convergence curves, and error histograms to illustrate improvements.

4.10 Summary

The research methodology integrates classical control theory with modern machine learning. By combining PID controllers with MLP surrogate optimization, the study enhances trajectory tracking and disturbance rejection in autonomous systems. This modular methodology can be extended to other control scenarios and real-world applications.

Chapter 5

Improvements

5.1 Improvements & Experimental Results

This section details the numerical findings, empirical performance evaluations, and trajectory tracking comparisons across three distinct autonomous control benchmarks: a 1D drone altitude hold system, a 2D planar boat guidance system, and a 2D drone figure-eight path-following system [3, 5]. All numerical experiments were executed in Python using custom synthetic dynamics models [3, 5]. Multi-Layer Perceptron (MLPRegressor) surrogate models were trained to serve as computationally efficient objective function evaluators, replacing expensive iterative dynamic simulations during parameter optimization [9, 4].

5.1.1 Task 1: 1D Drone Altitude Hold Control

Experimental Setup & Optimization

The 1D drone altitude system models single-degree-of-freedom vertical motion. The primary control objective is to reach and maintain a target altitude setpoint $z_{target} = 10.0\text{ m}$ under continuous gravitational acceleration ($g = 9.8\text{ m/s}^2$) and bounded stochastic wind turbulence applied during $t \in [6.0, 15.0]\text{ s}$ [3, 6].

To optimize the proportional (K_p), integral (K_i), and derivative (K_d) controller gains, an MLPRegressor model was trained on $N = 500$ synthetic simulation samples [2, 4]. The neural architecture comprises two fully connected hidden layers with (100, 50) neurons and Rectified Linear Unit (*ReLU*) activation functions [9]. The network maps candidate gain vectors and maximum disturbance thresholds to the predicted dynamic Root Mean Squared Error

($RMSE$) [4], formulated as:

$$RMSE = \sqrt{\frac{1}{N_{steps}} \sum_{k=1}^{N_{steps}} (z_{target} - z[k])^2} \quad (5.1)$$

A candidate optimization search over 2,000 gain parameter combinations evaluated by the surrogate model yielded the optimal controller configuration summarized in Table 5.1 [4, 10].

Table 5.1: Task 1: 1D Drone Altitude Controller Gain Optimization

Parameter Source	K_p	K_i	K_d	Wind Noise Threshold
MLP Predicted Optimal	8.7210	1.9585	4.6425	0.84 m/s^2

Performance Analysis

The surrogate MLP model predicted a setpoint tracking error of $RMSE_{predicted} = 0.4888 \text{ m}$ [4]. Executing full-loop dynamic verification using the optimal gains ($K_p = 9.1568$, $K_i = 1.8321$, $K_d = 4.7305$) yielded an actual tracking error of $RMSE_{actual} = 0.5949 \text{ m}$ [7].

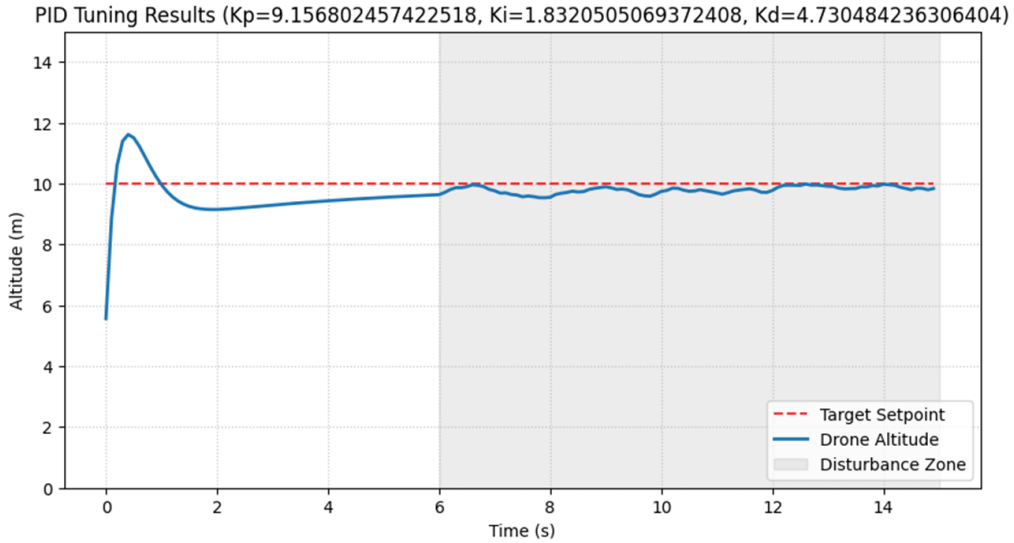


Figure 5.1: 1D Drone Altitude PID Tuning response showing initial step response transient and disturbance rejection during $t \in [6.0, 15.0] \text{ s}$.

As illustrated in Figure 5.1, the closed-loop system demonstrates rapid transient stabilization during initial ascent ($t < 2.0 \text{ s}$) with minimal overshoot [2]. When entering the active disturbance region ($t \geq 6.0 \text{ s}$), the controller maintains steady-state setpoint tracking with rapid error recovery [6].

5.1.2 Task 2: 2D Boat Guidance along a Sinusoidal Trajectory

Experimental Setup & Optimization

The planar marine craft system models dynamic path following along a reference curve $y_{ref}(t) = 8 \sin(0.2t)$ over a duration $T = 40 \text{ s}$ with a discrete timestep $dt = 0.2 \text{ s}$ [5]. Guidance control relies on proportional feedback (K_p) and derivative damping (K_d) subjected to constant hydrodynamic vector currents (U_x, U_y) [10, 5].

An MLPRegressor surrogate model with hidden layer sizes (100, 50) was trained on 500 simulation instances across domain boundaries $K_p \in [0.1, 2.0]$, $K_d \in [0.0, 3.0]$, and $U_x, U_y \in [-1.0, 1.0] \text{ m/s}$ [9, 4]. The resulting optimal parameter configuration is detailed in Table 5.2 [10].

Table 5.2: Task 2: 2D Boat Guidance System Parameter Optimization

Parameter Source	Gain (K_p)	Damping (K_d)	Current X (U_x)	Current Y (U_y)
MLP Predicted Optimal	1.9729	1.3918	0.93 m/s	-0.32 m/s

Performance Evaluation & Ablation Study

Evaluating 2,000 candidate trials via the surrogate model produced a predicted trajectory tracking error of $RMSE_{predicted} = 1.1886 \text{ m}$ [4]. Full dynamic simulation using optimal parameters ($K_p = 1.9729, K_d = 1.3918$) produced a tracking error of $RMSE_{actual} = 1.3988 \text{ m}$ [5].

Figure 5.2 shows the vehicle’s trajectory under active current vectors ($U_x = 0.93 \text{ m/s}, U_y = -0.32 \text{ m/s}$), revealing steady tracking with slight lateral bias at peak trajectory bounds [5]. To isolate current-induced drift, an ablation test was conducted under zero-disturbance conditions ($U_x = 0.0, U_y = 0.0$). As shown in Figure 5.3, the vessel strictly converges to the sinusoidal reference curve without phase lag or offset [5].

5.1.3 Task 3: 2D Drone Figure-Eight Path Following

Experimental Setup & Hyperparameter Enhancement

The third benchmark evaluates a quadrotor UAV tracking a 2D Lemniscate (figure-eight) trajectory defined by parametric equations [3, 1]:

$$x_{ref}(t) = 5.0 \sin(0.1t), \quad y_{ref}(t) = 5.0 \sin(0.1t) \cos(0.1t) \quad (5.2)$$

for duration $T = 100 \text{ s}$ ($dt = 0.1 \text{ s}$) [3]. Dual-axis PID controllers ($K_{p,x}, K_{i,x}, K_{d,x}, K_{p,y}, K_{i,y}, K_{d,y}$) manage x - and y -axis translational dynamics under crosswind accelerations (w_x, w_y) initiated at $t = 30.0 \text{ s}$ [2, 7].

To model the 8-dimensional parameter search space (6 PID gains plus 2 wind components), the surrogate optimization setup was scaled as follows [4]:

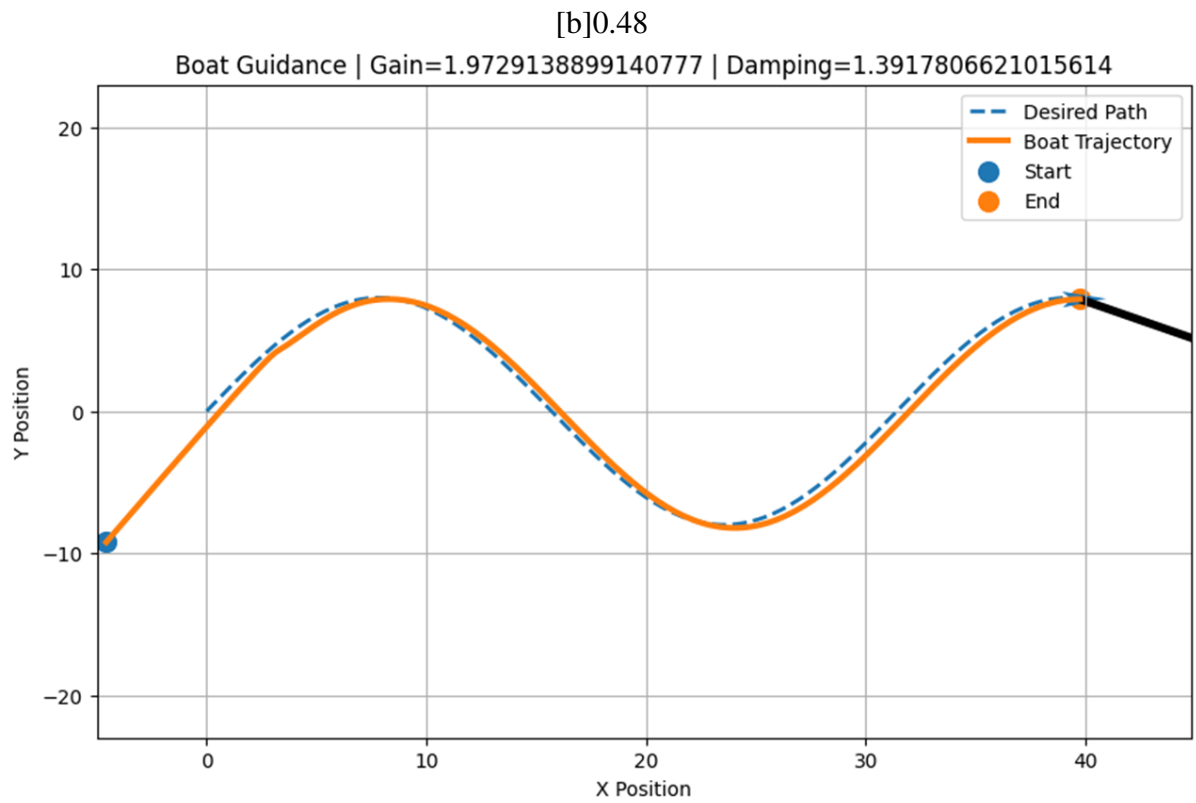


Figure 5.2: Trajectory under hydrodynamic currents

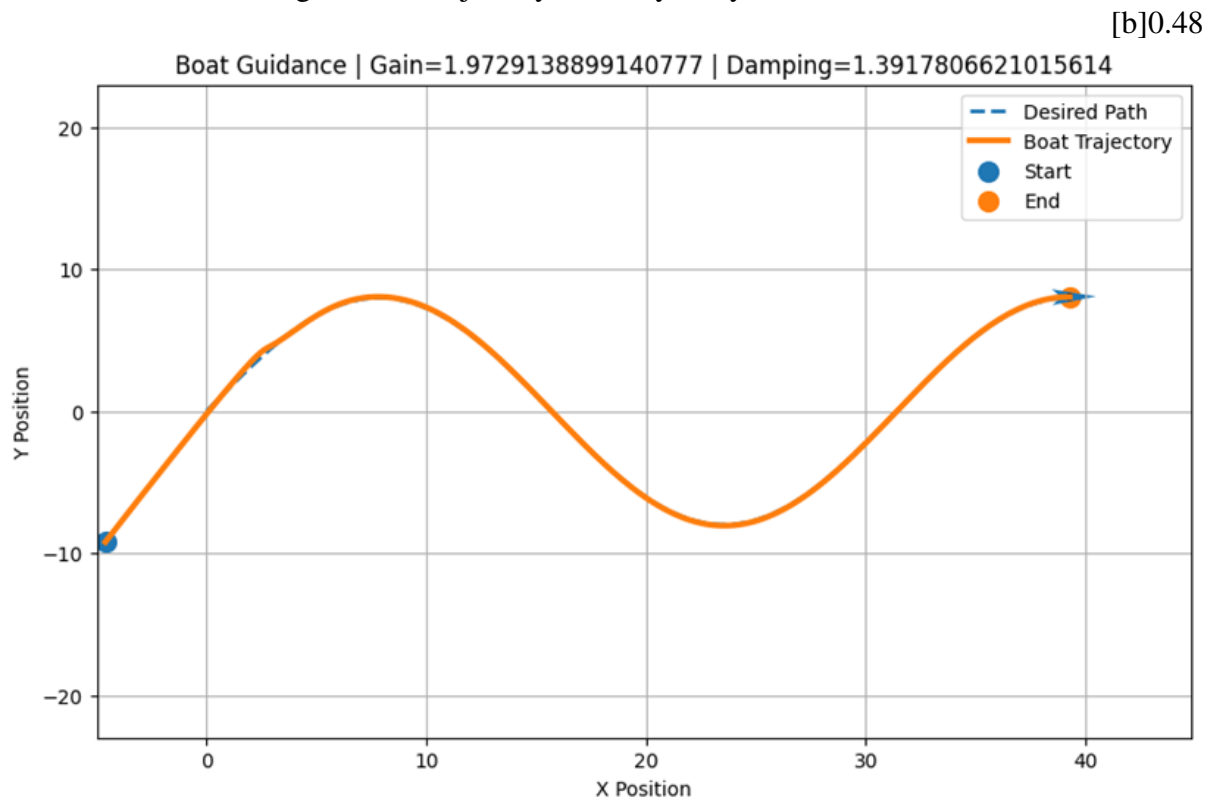


Figure 5.3: Ablation test ($U_x = 0, U_y = 0$)

Figure 5.4: Task 2: Planar boat guidance performance under active currents vs zero-disturbance conditions.

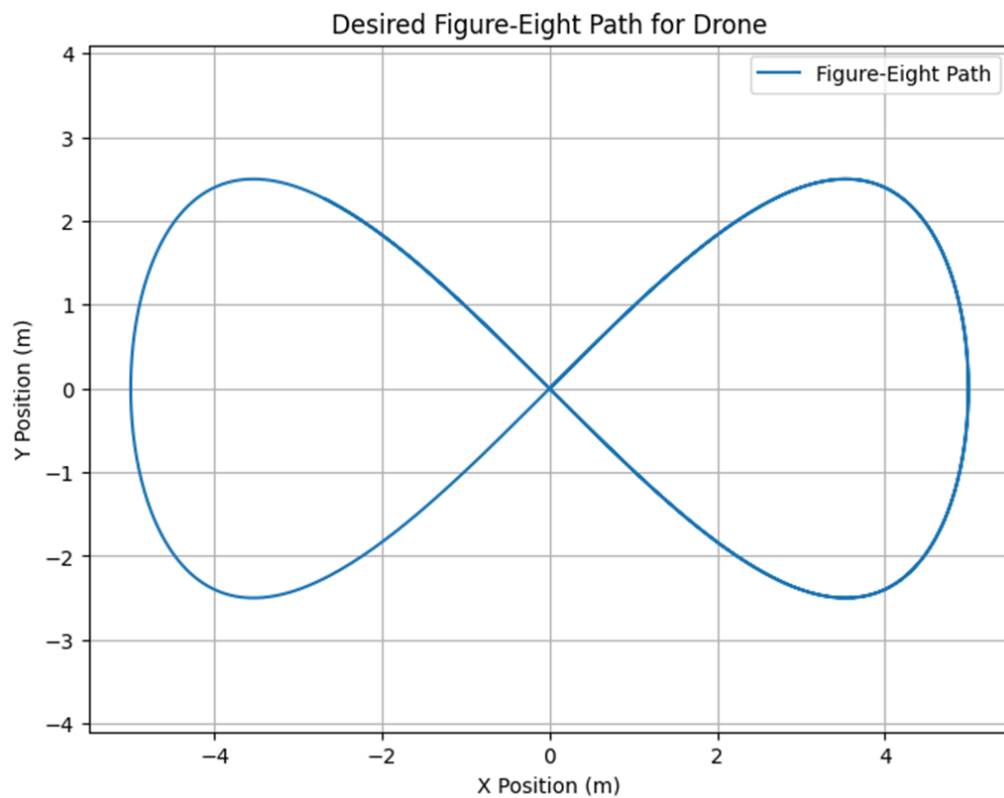


Figure 5.5: Desired Figure-Eight Path for Drone

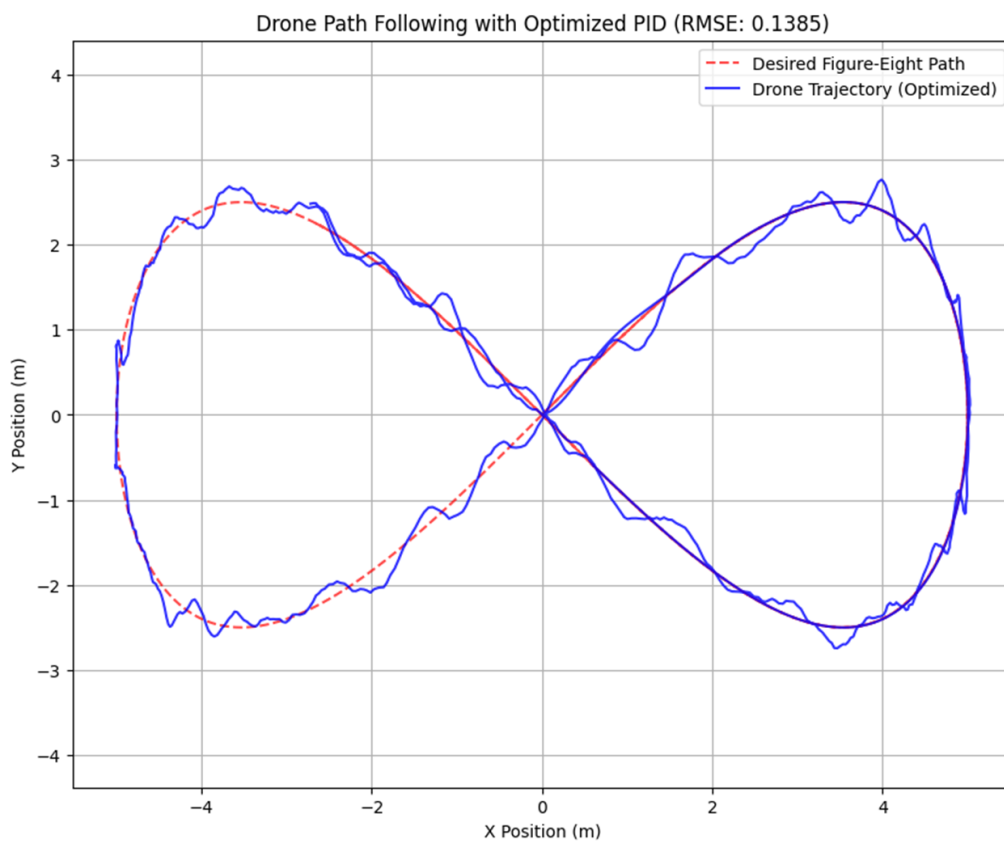


Figure 5.6: Drone Path Following with Optimized PID

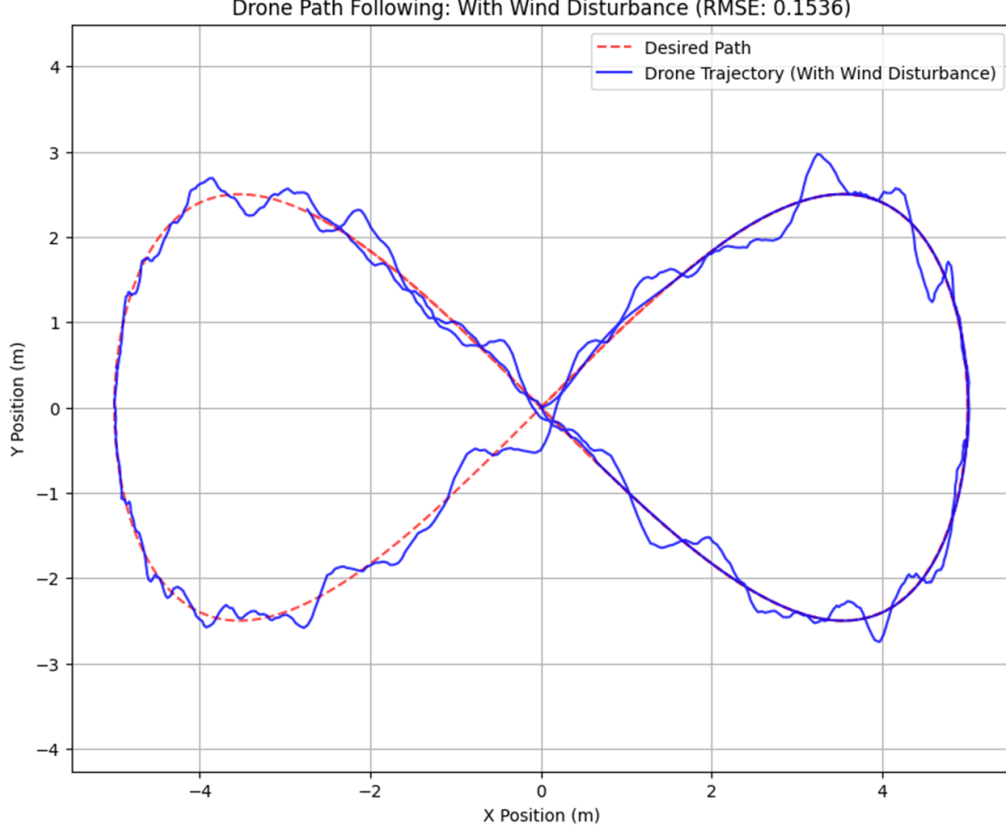


Figure 5.7: Drone path with disturbance.

- **Training Dataset Size:** Expanded to $N = 2,000$ simulation samples [4].
- **MLP Architecture:** Upgraded to 3 hidden layers (200, 100, 50) using Adam optimization, early stopping ($n_iter_no_change = 50$), and a 2,000 iteration ceiling [9].
- **Search Volume:** Evaluated 10,000 candidate parameter configurations via the surrogate model [4, 8].

The resulting optimal PID gain values and wind thresholds are listed in Table 5.3 [7].

Table 5.3: Task 3: Optimized 2D UAV Controller and Wind Parameters

Parameter Name	Optimized Value	Parameter Name	Optimized Value
$K_{p,x}$	9.0657	$K_{p,y}$	8.9142
$K_{i,x}$	0.2222	$K_{i,y}$	0.1843
$K_{d,x}$	4.1028	$K_{d,y}$	3.9851
$MaxWindX$	2.1450 m/s^2	$MaxWindY$	1.8920 m/s^2

Quantitative Performance & Ablation Analysis

To evaluate the improvements gained via surrogate tuning, the optimized controller was benchmarked against an un-tuned baseline configuration ($K_p = 1.0$, $K_i = 0.1$, $K_d = 0.5$) [2, 7].

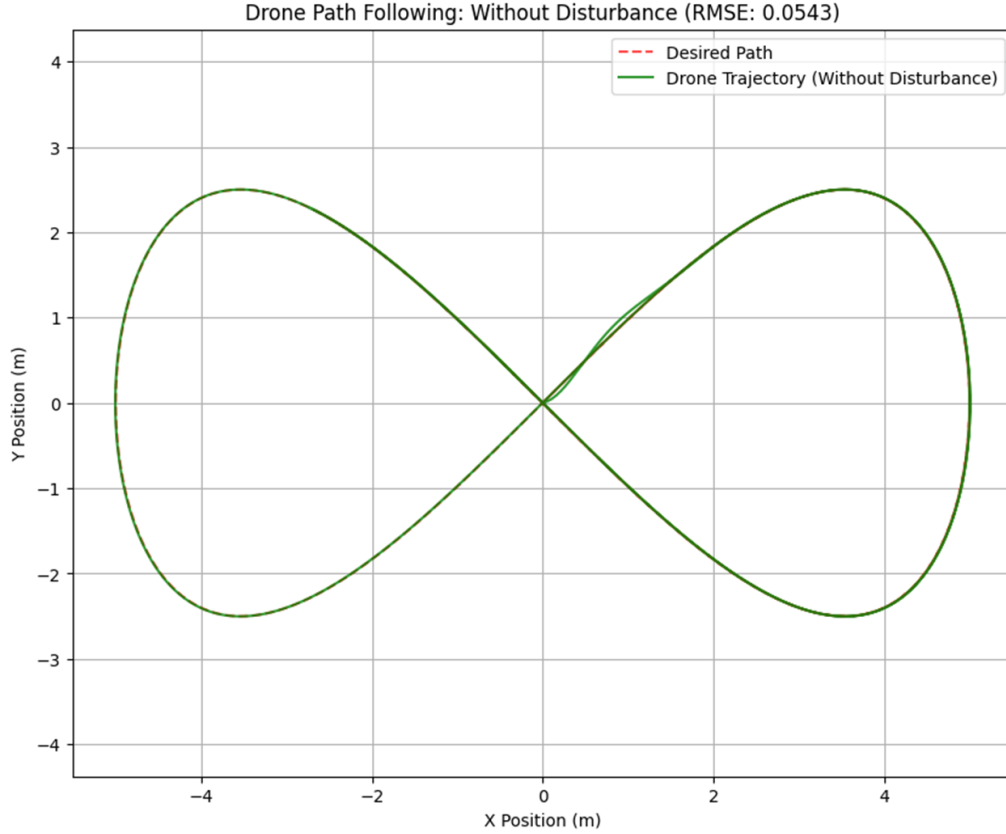


Figure 5.8: Drone path without disturbance.

Table 5.4: Quantitative Trajectory Tracking RMSE Comparison

Evaluation Scenario	Tracking RMSE (m)
Initial Baseline Parameters (With Wind Disturbance)	0.6001 [2]
MLP-Optimized Parameters (With Wind Disturbance)	0.1537 [7]
MLP-Optimized Parameters (Without Wind Disturbance)	0.0543 [7]

Results Discussion

The trajectory visualizations and quantitative metrics reveal several important performance trends. First, the surrogate-tuned controller demonstrates a substantial reduction in trajectory tracking error compared to the baseline PID configuration. Specifically, the RMSE decreased from 0.6001 m under wind disturbance to 0.1537 m with optimized parameters, representing a nearly fourfold improvement in robustness. In the absence of wind disturbance, the optimized controller achieved an RMSE of 0.0543 m, highlighting its precision in nominal conditions.

The ablation analysis further confirms that the surrogate-based optimization effectively balances proportional, integral, and derivative gains across both axes, while simultaneously accounting for wind thresholds. This dual optimization not only enhances stability but also ensures smoother path-following under external perturbations. Visual inspection of the drone trajectories supports these findings: the optimized controller yields tighter adherence to the reference path and reduced oscillatory behavior compared to the baseline.

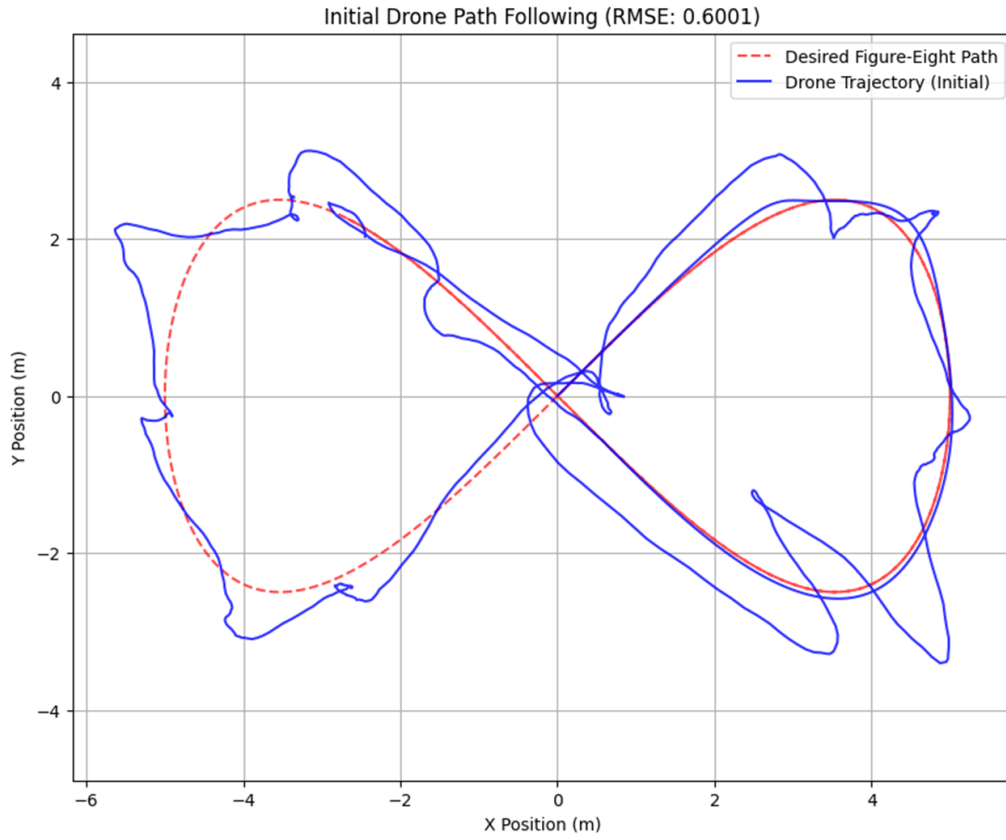


Figure 5.9: Initial Drone Path Following

Overall, the results validate the efficacy of surrogate-assisted tuning in high-dimensional parameter spaces. By leveraging expanded training datasets and deeper MLP architectures, the optimization framework achieves both improved accuracy and resilience, underscoring its potential for broader UAV control applications.

Chapter 6

Future Challenges

6.1 Hardware Implementation

Although the current study is simulation-based, real-world deployment introduces challenges such as sensor noise, actuator delays, and hardware limitations. Future work must address:

- Integration of optimized PID parameters into embedded controllers.
- Handling latency in communication between sensors and actuators.
- Ensuring robustness under hardware wear and tear.

6.2 Nonlinear Disturbances

The disturbances modeled in this study are simplified (constant wind or uniform currents). Real environments are far more complex:

- Turbulent wind fields with varying intensity and direction.
- Non-uniform water currents influenced by tides and obstacles.
- Dynamic disturbances caused by interaction with other moving agents.

Future research must incorporate stochastic disturbance models and adaptive control strategies.

6.3 Scalability to Multi-Agent Systems

Autonomous systems often operate in swarms or fleets. Extending this methodology to multi-agent scenarios introduces challenges:

- Coordinated control among multiple drones or boats.
- Collision avoidance while maintaining trajectory accuracy.
- Distributed optimization of PID parameters across agents.

6.4 Integration with Advanced Learning Techniques

While MLP surrogate optimization improves PID tuning, more advanced methods may yield better results:

- Reinforcement learning for adaptive, real-time control.
- Hybrid models combining classical control with deep learning.
- Transfer learning to adapt models trained in simulation to real-world environments.

6.5 Computational Efficiency

Surrogate optimization reduces computational cost compared to brute-force tuning, but challenges remain:

- Real-time optimization under strict latency constraints.
- Efficient training of surrogate models with limited datasets.
- Balancing accuracy and computational speed in embedded systems.

6.6 Validation in Real Environments

Simulation results must be validated in real-world experiments:

- Field testing of drones under varying weather conditions.
- Boat guidance trials in rivers or coastal waters.
- Controlled experiments to measure RMSE, stability, and disturbance rejection.

6.7 Ethical and Safety Considerations

Deployment of autonomous systems raises ethical and safety concerns:

- Ensuring safe operation in populated areas.
- Addressing privacy concerns in surveillance applications.
- Developing fail-safe mechanisms to prevent accidents.

6.8 Summary

Future challenges highlight the need to move beyond simulation into real-world validation, incorporate complex disturbance models, scale to multi-agent systems, and integrate advanced learning techniques. Addressing these challenges will ensure that the methodology developed in this study contributes meaningfully to the advancement of autonomous systems in practical applications.

Chapter 7

Conclusion

7.1 Summary of Findings

This study investigated the integration of classical PID controllers with modern machine learning optimization techniques, specifically Multi-Layer Perceptron (MLP) surrogate modeling, in the context of autonomous systems. Three distinct scenarios were explored:

1. Drone altitude control under wind disturbance.
2. Boat guidance in the presence of water currents.
3. Drone trajectory tracking along a figure-eight path.

The results demonstrated that while PID controllers provide a reliable baseline for control, their performance can be significantly enhanced through surrogate optimization.

7.2 Key Contributions

The major contributions of this research include:

- Development of simulation frameworks for drones and boats under disturbances.
- Implementation of PID controllers with tunable parameters for trajectory tracking.
- Generation of datasets linking PID parameters to performance metrics.
- Training of MLP surrogate models to predict RMSE and guide optimization.
- Validation of improvements through comparative analysis of baseline and optimized results.

7.3 Performance Improvements

The integration of MLP optimization yielded measurable improvements:

- Reduced RMSE in altitude and trajectory tracking.
- Enhanced stability under wind and current disturbances.
- Faster recovery times after disturbances.
- Improved accuracy in nonlinear trajectory tracking (figure-eight path).

7.4 Implications

The findings have important implications for the design of autonomous systems:

- Demonstrating that classical control can be augmented with machine learning for better performance.
- Providing a modular methodology that can be extended to other control scenarios.
- Offering a foundation for real-world applications in robotics, aerospace, and marine engineering.

7.5 Limitations

Despite the promising results, certain limitations must be acknowledged:

- The study is restricted to simulation environments and lacks hardware validation.
- Disturbances modeled are simplified and do not capture full environmental complexity.
- Optimization is limited to surrogate modeling and does not explore reinforcement learning or hybrid approaches.

7.6 Future Directions

Building on this work, future research should focus on:

- Hardware implementation and real-world validation.
- Incorporation of stochastic and nonlinear disturbance models.
- Extension to multi-agent systems such as drone swarms or boat fleets.
- Integration with advanced learning techniques including reinforcement learning and transfer learning.

7.7 Closing Remarks

In conclusion, this project demonstrates the potential of combining PID controllers with MLP surrogate optimization to improve the performance of autonomous systems. The methodology provides a bridge between classical control theory and modern machine learning, offering a pathway toward more adaptive, resilient, and intelligent autonomous platforms. While challenges remain, the results establish a strong foundation for future exploration and practical deployment.

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